

Zhaodong Liu

Email: zhaodongli@andrew.cmu.edu | Tel: (+1) 412-390-6409 | GitHub: github.com/zhaodong-liu
LinkedIn: [linkedin.com/in/zhaodong-liu](https://www.linkedin.com/in/zhaodong-liu)

EDUCATION

Carnegie Mellon University – Pittsburgh, PA Master of Computational Data Science (MCDS): Analytics Track	Dec 2027(Expected)
New York University Shanghai – Shanghai, China Bachelor of Science; Double Major: Computer Science, Mathematics, cum laude Cumulative GPA: 3.793/4.0 CS Major GPA: 3.946/4.0 Math Major GPA: 3.786/4.0	May 2026

SKILLS

Programming: Python, Java, C, SQL, R | **ML & AI:** PyTorch, LoRA/PEFT, Hugging Face, Transformers
Data & Tools: NumPy, Pandas, MySQL, Git | **Languages:** Chinese (native), English (fluent)

PROFESSIONAL EXPERIENCE

Data Engineer Intern, Data Strategy Department, McDonald's China, Shanghai, China	Dec 2025 - Apr 2026
- Engineered an internal AI agent for self-service data access, routing natural-language queries to constraint-aware SQL over a catalog spanning 14 business domains, or RAG over internal data maps and documents to generate cited answers. - Developed a similarity-based sales forecasting function over a combo master-data set normalized across 4 ID levels, quantifying menu-composition to forecast new-combo launch sales from comparable combos. - Managed and validated enterprise database schemas, governing a 203-scenario and 54-interface data map. - Redesigned data dashboards, organizing 500+ financial metrics with structured drill-down paths for self-service analytics.	

RESEARCH EXPERIENCE

Research Assistant, New York University Shanghai, Shanghai, China Generative Retrieval for Multimodal Recommendation System - Advisor: Prof. Hongyi Wen	May 2025 - Jan 2026
- Developed a multimodal framework for sequential recommendation on Amazon datasets. Identified and applied VQ-VAE, RQ-VAE, OPQ to capture discrete behavior patterns, achieving 47.2% improvement in NDCG@10 (0.0339 to 0.0499). - Implemented BPE-inspired partitioning tokenization across item sequences, enhancing coherence and representation quality.	
Research Assistant, New York University Shanghai, Shanghai, China Energy-Aware Route Optimization Under Traffic-Light Timing - Advisor: Prof. Zhibin Chen	May 2024 - Aug 2024
- Proposed and designed a traffic-aware routing system under real-world map constraints, minimizing energy consumption. - Built an OSM, SUMO, TraCI pipeline and independently extended Dijkstra's algorithm to multi-objective (time, energy) shortest paths, integrating electric and gasoline vehicle energy models. Achieved 10–15% fuel savings at 2–5% time cost.	

PROJECTS

Finetuning on Pretrained LM for Efficient Retrosynthesis of Chemical Reactions Center for Data Science, New York University, New York, NY	Sep 2024 - Dec 2024
- Trained predictive models for chemical reaction retrosynthesis using SMILES notation, applying an LLM (LlaSMol-Llama2) for dataset reconstruction and augmentation, improving Top-1 Accuracy by 26.2% (0.588 to 0.742). - Employed fine-tuning strategies (encoder freezing, MLP adapters, LoRA) to optimize training efficiency.	
Fantasy Sports League Database Tandon School of Engineering, New York University, Brooklyn, NY	Sep 2024 - Dec 2024
Self-directed a project to design a 13-table relational database in MySQL supporting multi-league sports with user authentication, draft scheduling, roster management, and trading systems. Engineered a Python backend and HTML/CSS frontend with ACID-compliant transactions, triggers, and stored procedures, integrating full-stack functionality.	

HONORS

New York University Shanghai, Founder's Day Award	May 2026
New York University Shanghai, Dean's List	Sep 2022 - May 2025
Mathematical Contest in Modeling, Certificate of Achievement, Honorable Mention	May 2023
Preprocessed large datasets and built predictive models, including Bayesian regression, for regional market data analysis.	

LEADERSHIP

Trombone Section Leader, Chamber Orchestra, New York University Shanghai	Sep 2022 - May 2026
Directed trombone section, coordinated weekly rehearsals and semester concerts, and oversaw instrument storage logistics.	
Treasurer, PEER-Pal Club, New York University Shanghai	May 2023 - Aug 2024
Managed financial operations and budgeting; facilitated letter exchanges with high school students from Hunan Province.	